

Week 6: SSH Key Authentication (Part 2)

Date: April 13, 2026

Study Time: 5-8 hours total

GOAL

Complete SSH key deployment across entire infrastructure

STUDY TASKS (1-2 hours)

Reading

- ☐ "How Linux Works" by Brian Ward
 - Chapter 10: SSH hardening and security

Video Learning

- ☐ LearnLinuxTV - Search: "SSH Hardening Security"
 - <https://youtube.com/@LearnLinuxTV>
- ☐ Professor Messer - Search: "SSH Security"
 - <https://youtube.com/@professormesser>

Concepts to Understand

- ☐ Disabling password authentication
 - ☐ Disabling root login
 - ☐ Changing default SSH port
 - ☐ SSH config file (`~/.ssh/config`)
 - ☐ fail2ban for brute force protection
 - ☐ SSH tunneling basics
-

LAB WORK (3-4 hours)

Harden SSH Server Configuration

- ☐ Backup sshd_config first!

```
bash
```

```
sudo cp /etc/ssh/sshd_config /etc/ssh/sshd_config.backup
```

☐ Edit /etc/ssh/sshd_config

```
bash
```

```
sudo nano /etc/ssh/sshd_config
```

```
# Find and modify these lines:
```

☐ Disable password authentication

```
# Find this line:
```

```
#PasswordAuthentication yes
```

```
# Change to:
```

```
PasswordAuthentication no
```

```
# This forces key-only authentication
```

☐ Disable root login

```
# Find:
```

```
#PermitRootLogin yes
```

```
# Change to:
```

```
PermitRootLogin no
```

```
# Root can't SSH in directly
```

```
# Must SSH as user, then sudo
```

☐ Change SSH port (optional)

```
# Find:
```

```
#Port 22
```

```
# Change to something like:
```

```
Port 2222
```

```
# Remember to update firewall rules!
```

```
# Less automated attacks on non-standard ports
```

☐ **Other recommended settings**

```
# Only allow specific users
AllowUsers your_username tier1admin

# Or allow groups
AllowGroups ssh-users

# Limit auth attempts
MaxAuthTries 3

# Timeout for login
LoginGraceTime 30

# Disable empty passwords
PermitEmptyPasswords no

# Disable X11 forwarding if not needed
X11Forwarding no
```

Restart SSH Service

☐ **Check config syntax**

```
bash

sudo sshd -t
# Should return nothing if config is valid
```

☐ **Restart SSH daemon**

```
bash

# Ubuntu:
sudo systemctl restart ssh

# Fedora/RHEL:
sudo systemctl restart sshd

# Verify it's running:
sudo systemctl status sshd
```

☐ **IMPORTANT: Test before closing current session!**

```
bash
```

```
# In a NEW terminal window/tab:
```

```
ssh user@server
```

```
# Make sure you can still connect!
```

```
# If not, you can fix it in your original session
```

Create ~/.ssh/config for Easy Connections

☐ Create client config file

```
bash
```

```
nano ~/.ssh/config
```

☐ Add host shortcuts

```
# Fedora Server
Host fedora
  HostName 192.168.1.100
  User your_username
  Port 2222
  IdentityFile ~/.ssh/id_ed25519
```

```
# Raspberry Pi 4 (Tier 4)
Host pi4
  HostName 192.168.1.101
  User pi
  IdentityFile ~/.ssh/id_ed25519
```

```
# Raspberry Pi Zero (Tier 3)
Host pi-zero
  HostName 192.168.1.102
  User pi
  IdentityFile ~/.ssh/id_ed25519
```

☐ Set correct permissions

```
bash
```

```
chmod 600 ~/.ssh/config
```

☐ Test shortcuts

```
bash
```

```
# Now you can just type:
```

```
ssh fedora
```

```
ssh pi4
```

```
ssh pi-zero
```

```
# Instead of full commands!
```

Deploy Keys to All Pi Devices

☐ Raspberry Pi 4 (Tier 4)

```
bash
```

```
ssh-copy-id pi@pi4-ip-address
```

```
ssh pi4 # Test
```

☐ Raspberry Pi Zero (Tier 3)

```
bash
```

```
ssh-copy-id pi@pi-zero-ip-address
```

```
ssh pi-zero # Test
```

☐ Harden each Pi

```
bash
```

```
# SSH into each Pi:
```

```
ssh pi4
```

```
# Edit SSH config:
```

```
sudo nano /etc/ssh/sshd_config
```

```
# Apply same hardening as above
```

```
# Restart SSH:
```

```
sudo systemctl restart ssh
```

```
# Test in new terminal before closing!
```

Update Firewall Rules

☐ If you changed SSH port to 2222

bash

```
# Ubuntu (UFW):  
sudo ufw allow 2222/tcp  
sudo ufw delete allow 22/tcp  
sudo ufw reload  
  
# Fedora (firewalld):  
sudo firewall-cmd --permanent --add-port=2222/tcp  
sudo firewall-cmd --permanent --remove-service=ssh  
sudo firewall-cmd --reload
```

FLASHCARD COMMANDS

Create flashcards for these:

1. `sshd_config`
2. `systemctl restart sshd`
3. `ssh -i` (specify identity file)
4. `~/.ssh/config`
5. `ssh -p` (specify port)
6. `sshd -t` (test config)

Security Concepts:

- Why disable password auth?
- Why disable root login?
- Pros/cons of changing SSH port

EXAM OBJECTIVES

CompTIA Linux+ (XK0-005) Exam Objectives: [https://partners.comptia.org/docs/default-source/resources/comptia-linux-xk0-005-exam-objectives-\(3-0\)](https://partners.comptia.org/docs/default-source/resources/comptia-linux-xk0-005-exam-objectives-(3-0))

Relevant Sections:

- 3.2 Given a scenario, implement identity management
- 3.3 Given a scenario, implement and configure Linux firewalls
- 3.4 Given a scenario, implement and execute security best practices

QUARTERLY CHECK-IN #1

Time to test your knowledge! Message me: "Quiz me on Weeks 1-6"

Topics covered so far:

- File permissions (basic & special)
- User & group management
- sudo configuration
- SSH key authentication & hardening

Self-assessment:

- What's your confidence level (1-10)? _____
- Which topics need more review? _____
- Ready to move to systemd? _____

NOTES & REFLECTIONS

SSH deployment checklist:

- ☐ Laptop → Fedora (keys deployed, hardened)
- ☐ Laptop → Pi 4 (keys deployed, hardened)
- ☐ Laptop → Pi Zero (keys deployed, hardened)
- ☐ Desktop → All servers (keys deployed)
- ☐ iPhone → All servers (keys configured in Termius)

SSH config shortcuts created:

Port changes made:

Issues encountered and resolved:

Completed: ☐

QUICK REFERENCE

~/.ssh/config Example:

```
Host *
    ServerAliveInterval 60
    ServerAliveCountMax 3

Host shortname
    HostName actual-server.com
    User username
    Port 2222
    IdentityFile ~/.ssh/id_ed25519
```

sshd_config Security Checklist:

- ☒ PasswordAuthentication no
- ☒ PermitRootLogin no
- ☒ Port changed (optional)
- ☒ AllowUsers or AllowGroups set
- ☒ MaxAuthTries 3
- ☒ PermitEmptyPasswords no

Testing Checklist:

```
bash

# 1. Check config syntax
sudo sshd -t

# 2. Restart service
sudo systemctl restart sshd

# 3. Test in NEW terminal (don't close old one!)
ssh user@server

# 4. If it works, close old terminal
# 5. If it doesn't, fix in old terminal!
```

Emergency Recovery: If you lock yourself out:

1. Physical access to machine
 2. Revert to backup: `sudo cp /etc/ssh/sshd_config.backup /etc/ssh/sshd_config`
 3. Restart SSH: `sudo systemctl restart sshd`
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Previous Week: SSH Key Basics

Next Week: Systemd Service Management